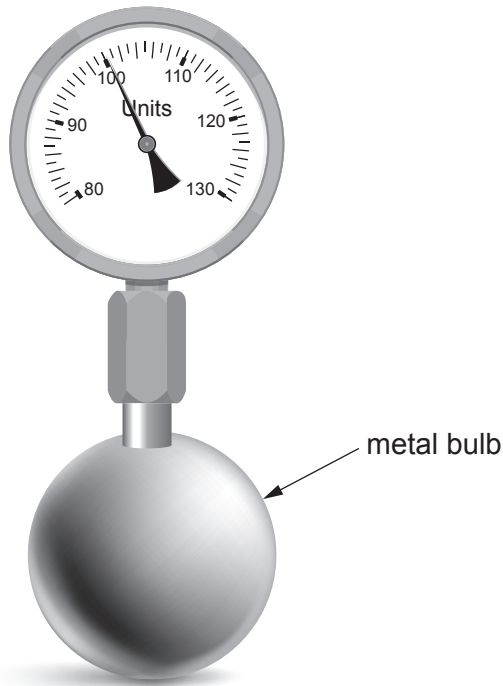


7. A teacher uses the apparatus shown below to demonstrate the link between the temperature and pressure of a gas.

The metal bulb, which is attached to a pressure gauge, contains a fixed volume of air.

The bulb is placed in a water bath and measurements of pressure for different temperatures are recorded.



- (a) State the independent variable in this experiment.

[1]

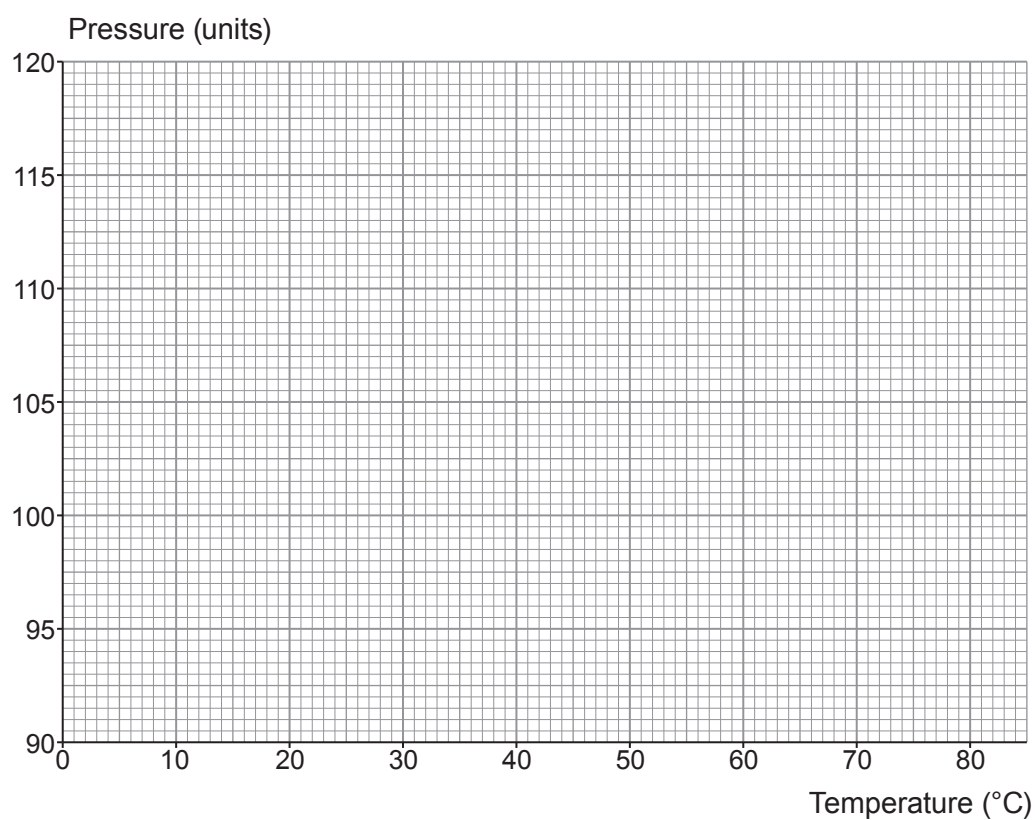
The following results were obtained.

Temperature (°C)	Pressure (units)
0	94
20	100
40	106
60	112
80	118



- (b) (i) Plot the data on the grid below and draw a suitable line.

[3]



- (ii) Describe the relationship between the temperature and the pressure of the gas.

[2]

.....

.....



- (iii) I. Use information from the graph or the table to determine the decrease in pressure for every **10 °C** decrease in temperature. [2]

decrease in pressure = units

- II. Use your answer to part I. above to determine the pressure of the gas at **−10 °C**. [1]

pressure = units

- (iv) State the value of absolute zero in °C. [1]

absolute zero = °C

- (c) The specific heat capacity of air, c , at constant volume is 720 J/kg °C.

Use the equation:

heat energy supplied = mass $\times$ specific heat capacity $\times$ change in temperature

$$Q = mc\Delta\theta$$

to calculate the heat energy required to raise the temperature of 0.05 kg of air **from** 20 °C **to** 60 °C. [2]

heat energy = J

